SHAF HAIDER

Geoinformatics Engineer | Remote Sensing Expert | Geospatial Software Developer
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Professional Summary

GIS and remote sensing specialist with 3 year of professional experience delivering national-scale Earth-observation analysis, spatial data pipelines, WebGIS, and field-mapping solutions. Proficient in Google Earth Engine, QGIS, ArcGIS, Python/GDAL, and PostGIS, with hands-on experience in multi-temporal land-cover classification, change detection, raster QA/QC, and geospatial service deployment. Currently **leads geospatial work for UNCCD 2026 national reporting** and served as **one of five named lead authors on a UNOPS-supported forest-carbon study**.

Technical Skills

Remote Sensing & Earth Observation: Google Earth Engine, Sentinel-2, Landsat, MODIS, multi-temporal LULC classification, Random Forest, change detection, spectral unmixing, canopy-density assessment, accuracy assessment, and QA/QC.
GIS & Spatial Analysis: QGIS, ArcGIS/ArcGIS Enterprise, ArcGIS Online, GRASS GIS, GDAL, GeoPandas, PostGIS, spatial databases, AHP, spatial statistics, and Open Foris Collect Earth.
WebGIS & Data Services: GeoServer, MapServer, GeoNode, Mapbox GL, Leaflet, Cesium, OpenLayers, Kepler.gl, MBTiles, tile servers, React, Next.js, FastAPI, Django REST Framework, and Node.js/Express.
Programming & Platforms: Python, JavaScript/TypeScript, R, C++, Kotlin, scikit-learn, PyTorch, TensorFlow, OpenCV, AWS, Docker, Kubernetes, MongoDB, and Git.

Professional Experience

GISPLUS TOTAL SOLUTIONS (Pvt.) Ltd. **June 2025-Present**
GIS & RS Expert *Islamabad, Pakistan*

UNOPS-Supported Forest-Carbon Monitoring Study
One of five named lead authors | Responsible for complete geospatial activity-data development | UNOPS-supported study

- Developed the complete activity-data workflow for a UNOPS-supported forest-carbon study and served as one of five named lead authors: processed Sentinel-2 imagery for 2016, 2020, and 2024 in Google Earth Engine and QGIS to produce 10 m LULC datasets using Random Forest classification and interpretation-based cross-year refinement.
- Quantified land-cover transitions, forest disturbance, and canopy-density change across four jurisdictions using change analysis and spectral unmixing: integrated outputs with field inventories and IPCC-aligned carbon accounting and supported uncertainty assessment, field verification, and stakeholder validation

UNCCD 2026 National Reporting Support for IUCN Pakistan
GIS Expert and Geospatial Team Lead

- Lead the GIS and remote-sensing workstream for Pakistan's UNCCD 2026 national reporting with IUCN Pakistan, coordinating national assessments of land degradation, Land Degradation Neutrality, SDG Indicator 15.3.1, and drought while guiding data review, QA/QC, and technical documentation.
- Participate and delivered provincial workshops for government and technical stakeholders on Earth-observation and Satellite based land degradation assessment and the Trends.Earth workflow.

National Forest Monitoring System (NFMS) (Platform Link)
Geospatial Data Engineering and WebGIS Development

- Contributed as part of the development team to building Pakistan’s National Forest Monitoring System in Next.js, an operational platform for accessing and interpreting forest-monitoring, land-cover, and climate-reporting information.

- Engineered a Python/GDAL raster pipeline for Pakistan's National Forest Monitoring System to harmonize inputs, automate QA/QC, create optimized MBTiles, publish tile services, and integrate large Earth-observation layers into a responsive Next.js WebGIS.

GeoFasal Agri tech Platform (Platform Link)
GISPlus Selected Assignments

- Developed field-level geospatial workflows for soil-suitability assessment, crop recommendations, and location-specific weather and agricultural guidance through an interactive mapping platform.
- Integrated Sentinel satellite products through STAC APIs, automated NDVI and NDWI computation for user-defined fields and used the derived indicators as inputs to machine-learning models for crop-yield prediction.

WebGIS for Spatial Property Intelligence
As a part of GISPlus Development team I contributed to the geospatial functionality for two production Next.js platforms serving DHA Islamabad–Rawalpindi and DHA Lahore. Integrated interactive mapping and spatial property discovery with verified records, seller dashboards, and realtor-management workflows.

Final Year Design Project

MOVE SMART: Climate-Aware Routing & Spatial Decision Support ***2024 - 2025***

- Co-developed an end-to-end decision-support system combining satellite imagery, street-level observations, cadastral data, and OpenStreetMap networks to generate pedestrian routes based on heat exposure, shade, connectivity, and accessibility.
- Built a Google Earth Engine workflow with Landsat, Sentinel-2, and MODIS to derive vegetation, canopy, and land-surface-temperature indicators; identified urban heat-island intensity up to 12 C and an inverse vegetation-temperature relationship (r = -0.72), then calculated road-segment comfort scores using AHP and PostGIS.
- Fine-tuned a tree-trunk detection model on 450 annotated images, achieving approximately 75% mAP and detecting 1,200+ trees; integrated the results with customized GraphHopper routing and 5-30-minute isochrones in a Next.js, Mapbox GL, FastAPI, and PostGIS platform.

Upwork Top Rasted Freelancer: GIS Expert & Full-Stack GIS Developer

profile link: *Shaf Haider* ***June 2021 - Present***

MOTTAINAI RECYCLING, NIGERIA (Organization Link**)**
Kotlin Android App Specialist and GIS Expert

- Administered Mottainai's ArcGIS Online organization and managed the operational geospatial data and services supporting customer pickups, driver dispatch, and field-management workflows.
- Developed and deployed three Kotlin-based Android applications using ArcGIS Maps SDK, ArcGIS Online services, Mapbox SDK, and device GPS. Implemented offline location capture and synchronization for reliable field mapping and data collection in low-connectivity areas.
- Designed a custom GraphHopper multi-stop routing solution that optimized pickup sequences for recycling trucks and navigation for field managers, reducing reliance on paid Google Maps and Mapbox routing services and helps **cutting recurring routing costs for the company by at least 28%.**

Live Android applications on Google Play: Mottainai – Waste & Recycling · Mottainai Field Manager · Mottainai Driver

Internships Experience

1. JUGRAFIYA (Pvt.) Ltd. (Company Profile Link**)** **Rawalpindi, Pakistan**
Web GIS Intern Full-Stack Geospatial Development *Jun 2024–Sep 2024*

- Led a three-developer team in building geospatial web applications with React and the vis.gl ecosystem, including deck.gl, Kepler.gl, and loaders.gl, for interactive 2D and 3D spatial visualization.
- Researched and implemented efficient browser-rendering workflows for large point-cloud, GeoJSON, and raster datasets used in client-facing applications.
- Developed a routing and geofencing application using OpenStreetMap and GraphHopper, generating alerts for routes intersecting high-traffic and high-crime zones.

2. INSTITUTE OF GEOGRAPHICAL INFORMATION SYSTEMS (IGIS), NUST
*GIS Research Intern - **NCGSA-funded Dengue/Zika Vector Monitoring Project*** *July - Aug 2023*

- Developed WebGIS and computer-vision components for an IoT research project focused on invasive Dengue/Zika vector monitoring and outbreak risk.
- Configured GeoServer-based spatial services and applied OpenCV in image-processing workflows during an intensive six-week internship.

Education

National University of Science and Technology (NUST) **2021-2025**
Bachelor of Geoinformatics Engineering, School of Civil and Environmental Engineering *Islamabad, Pakistan*
CGPA: 3.49/4.00
Final-Year Design Project: *Urban Mobility and Urban Microclimate: A Live Tool for Best Urban Routes in the Wake of Growing Urban Heat Island Phenomena*
Received **A grades** in **Final-Year Project I** and **II**.
Relevant Coursework: Remote Sensing; GIS Programming; Spatial Data Analysis; Digital Mapping and Image Processing; Machine Learning; Spatial Databases; Big Data Analytics; Spatial Decision Support Systems.